

Some of these models of concurrency are primarily intended to support reasoning and specification, while others can be used through the entire development cycle, including design, implementation, proof, testing, and simulation of concurrent systems. Some of these are based on message passing, while others have different mechanisms for concurrency.

The proliferation of different models of concurrency has motivated some researchers to develop ways to unify these different theoretical models. For example, Lee and Sangiovanni-Vincentelli have demonstrated that a so-called "tagged-signal" model can be used to provide a common framework for defining the denotational semantics of a variety of different models of concurrency,[11] while Nielsen, Sassone, and Winskel have demonstrated that category theory can be used to provide a similar unified understanding of different models.[12]

The Concurrency Representation Theorem in the actor model provides a fairly general way to represent concurrent systems that are closed in the sense that they do not receive communications from outside. (Other concurrency systems, e.g., process calculi can be modeled in the actor model using a two-phase commit protocol.[13]) The mathematical denotation denoted by a closed system S is constructed increasingly better approximations from an initial behavior called S using a behavior approximating function progression_S to construct a denotation (meaning) for S as follows:[14]